

Assignment - III

OFFICE NETWORK CONFIGURATION

Descriptive Technical Report

Cisco Packet Tracer Office Network with Backup and Recovery Simulation

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1. Introduction

The purpose of this practical exercise was to design and document a small office network with basic services and a simple recovery strategy. The simulated environment contains three client computers, one server, one network printer, and one Layer 2 switch. Static IPv4 addressing was assigned so that every device could be identified consistently. Connectivity was then tested using ICMP ping, and the server was configured as a DNS server so that the name office-server could resolve to the server IP address. A simple FTP file service was enabled to represent shared-file access. A backup and recovery event was simulated by downloading a server file to a client, deleting the original server copy, and uploading the backup copy back to the server.

Platform	Cisco Packet Tracer on Windows 11
Network	192.168.1.0/24
Devices	3 clients, 1 server, 1 printer, 1 switch
Objective	Simulate an office network, name resolution, file service, backup, failure, and recovery

Table 1 Descriptive

This report describes the Packet Tracer simulation that was completed. The assignment mentions Samba/Windows file sharing and rsync/robocopy. Packet Tracer does not provide those operating-system tools, so FTP was used as the file-service and backup/restore simulation. This is explicitly identified rather than presented as a real Samba/rsync deployment.

2. Network Topology

The final topology uses a single 2960-24TT switch as the central connection point. PC1, PC2, PC3, Server0, and Printer0 are connected to separate switch ports using copper straight-through Ethernet links. Because all five end devices are in the same IPv4 subnet, they can communicate directly at Layer 2 without requiring a router or default gateway.

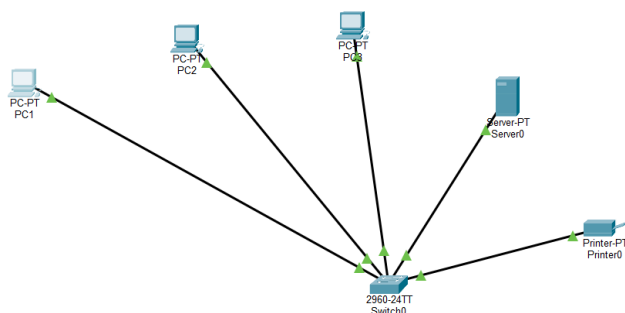


Figure 1 network topology

3. IP Addressing and Subnetting

The network was designed as 192.168.1.0/24. The subnet mask 255.255.255.0 means that the first three octets identify the network and the last octet is used for host addressing. No default gateway was required for the local, single-subnet tests performed in this simulation.

Device	Role	IPv4 Address	Subnet Mask
Server0	Server / DNS / FTP	192.168.1.10	255.255.255.0
PC1	Client 1	192.168.1.11	255.255.255.0
PC2	Client 2	192.168.1.12	255.255.255.0
PC3	Client 3	192.168.1.13	255.255.255.0
Printer0	Network printer	192.168.1.20	255.255.255.0

Table 2 Specification

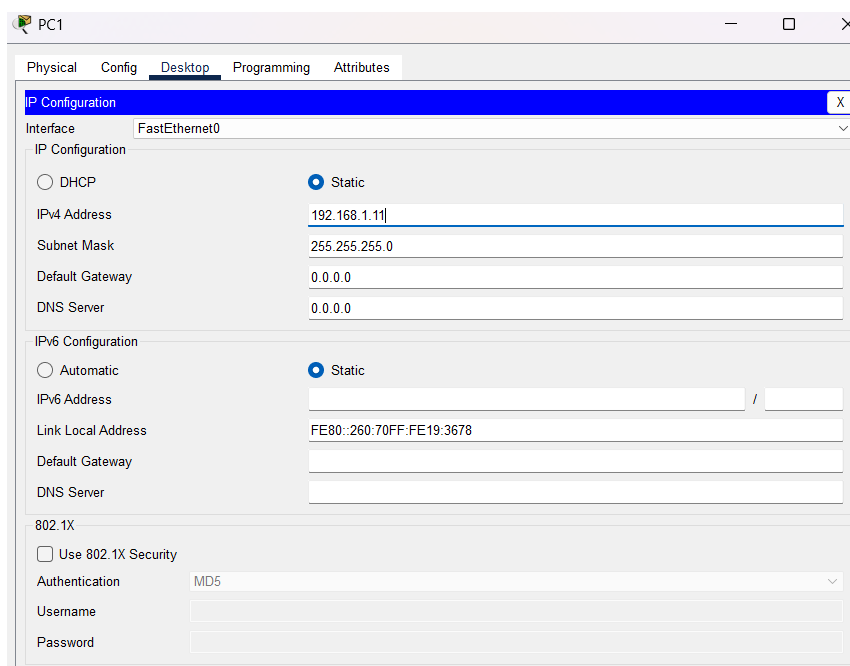


Figure 2 PC1 IP config

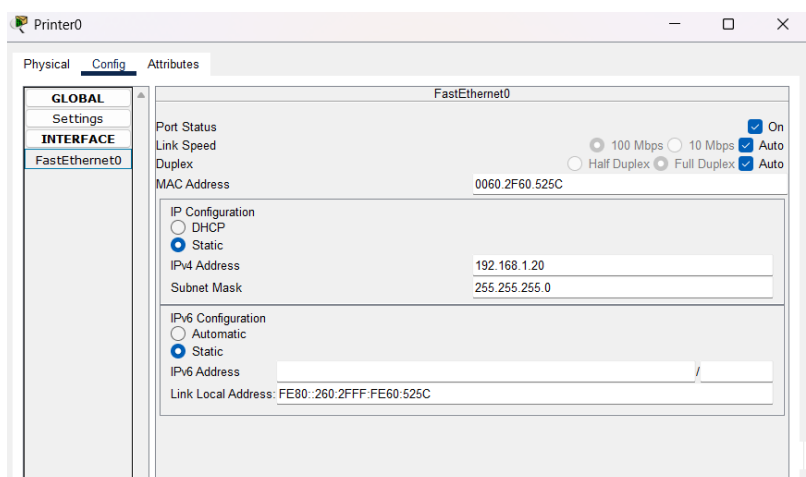


Figure 3 Printer IP config

4. Connectivity Testing

Connectivity was tested from PC1 to Server0 using ICMP. The command ping 192.168.1.10 returned four successful replies. The observed result was 4 packets sent, 4 packets received, and 0% packet loss, demonstrating that the client and server were reachable across the switch.

```
ping 192.168.1.10
Reply from 192.168.1.10: ...
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

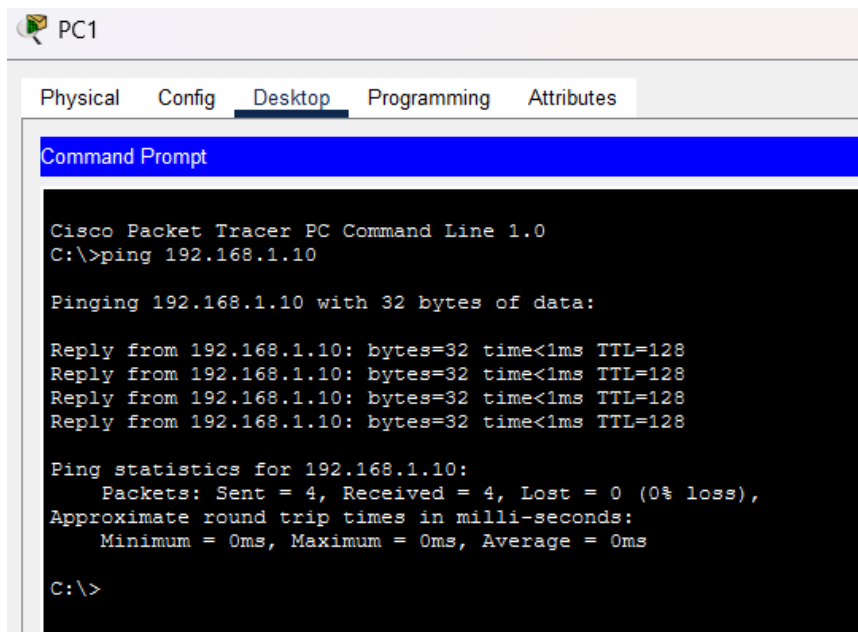


Figure 4 ping PC1

5. DNS Configuration and Name Resolution

Server0 was configured to provide DNS service. An A record named office-server was created and mapped to 192.168.1.10. PC1 was then configured to use 192.168.1.10 as its DNS server. A second ping test used the hostname office-server rather than the numeric IP address. The hostname resolved to 192.168.1.10 and produced successful replies, showing that DNS name resolution worked in the simulation.

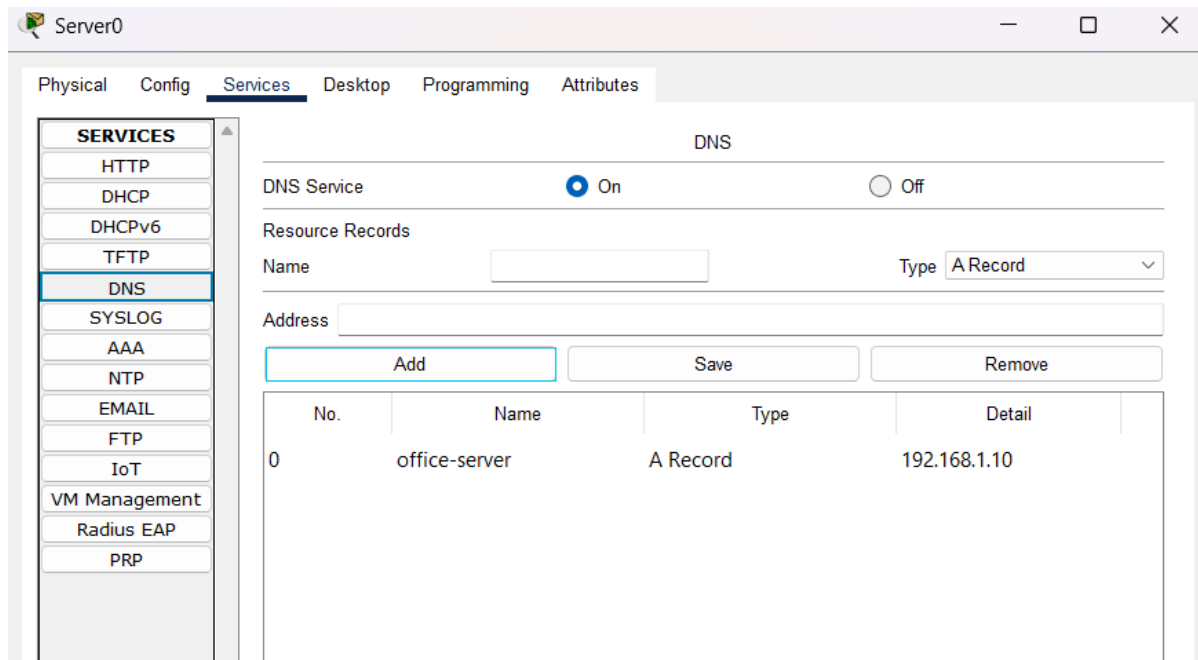


Figure 5 DNS config

6. File-Service Configuration

The assignment calls for shared folders and a printer through a file server using Samba or Windows File Sharing. Packet Tracer does not reproduce a full Windows/SMB or Linux/Samba environment. To keep the network simulation operational within the available time, FTP service on Server0 was enabled and an office user account was created with read, write, delete, rename, and list permissions. This provides a concrete file-service test inside Packet Tracer while avoiding an inaccurate claim that Samba or SMB was installed.

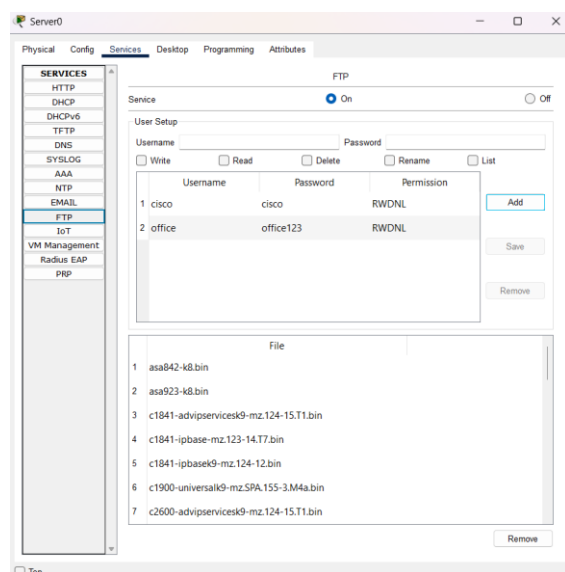


Figure 6 Server0 FTP service enabled with the office user account and full file permissions.

```

C:\>ftp 192.168.1.10
Trying to connect...192.168.1.10
Connected to 192.168.1.10
220- Welcome to PT Ftp server
Username:office
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.1.10:
 0 : asa842-k8.bin                    5571584
 1 : asa923-k8.bin                    30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 3 : c1841-ipbase-mz.123-14.T7.bin    13832032
 4 : c1841-ipbasek9-mz.124-12.bin     16599160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 7 : c2600-i-mz.122-28.bin           5571584
 8 : c2600-ipbasek9-mz.124-8.bin      13169700
 9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin  5571584
12 : c2800nm-ipbasek9-mz.124-8.bin    15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2950-i6q412-mz.121-22.EA4.bin  3058048
15 : c2950-i6q412-mz.121-22.EA8.bin  3117390
16 : c2960-lanbase-mz.122-25.FX.bin  4414921
17 : c2960-lanbase-mz.122-25.SEE1.bin 4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin 4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : ie9k_iosxe.17.09.04.SPA.bin     596133776
27 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
28 : ir800-universalk9-mz.SPA.155-3.M 61750062
29 : ir800-universalk9-mz.SPA.156-3.M 63753767
30 : ir800_yocto-1.7.2.tar           2877440
31 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
32 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin          5571584
34 : pt3000-i6q412-mz.121-22.EA4.bin  3117390
ftp>

```

Figure 7 ftp login & dir

7. Backup Strategy

The backup strategy used in this simulation is a simple client-side copy of a server file. The selected file was c2960-lanbase-mz.122-25.FX.bin. PC1 connected to the FTP server and downloaded the file using the FTP get command. The transfer completed successfully with a size of 4,414,921 bytes. The downloaded copy was treated as the backup copy for the failure/recovery test.

- Source: Server0 FTP file service at 192.168.1.10.
- Backup destination: PC1 local storage within the Packet Tracer simulation.
- Backup object: c2960-lanbase-mz.122-25.FX.bin.
- Recorded size: 4,414,921 bytes.
- Integrity check used in the exercise: the restored file reappeared in the FTP directory listing after upload.

Real office implementation recommendation use scheduled rsync on Linux or robocopy on Windows, store backups on a separate storage target, retain multiple versions, and periodically test restoration. The present report only claims the FTP simulation that was executed.

```

ftp>get c2960-lanbase-mz.122-25.FX.bin

Reading file c2960-lanbase-mz.122-25.FX.bin from 192.168.1.10:
File transfer in progress...

[Transfer complete - 4414921 bytes]

4414921 bytes copied in 38.548 secs (3111 bytes/sec)
ftp>

```

Figure 8 ftp backup

8. Failure Simulation and Recovery

To test recoverability, the original server-side file was deliberately deleted from the FTP directory. A subsequent directory listing showed that the selected file was no longer present, representing a simple file-loss incident. The backup copy stored on PC1 was then uploaded back to Server0 using the FTP put command. The transfer completed successfully with the same recorded size of 4,414,921 bytes. A final directory listing showed that c2960-lanbase-mz.122-25.FX.bin had returned to the server, verifying recovery.

Stage	Action	Result
1. Backup	FTP get from Server0 to PC1	4,414,921-byte transfer completed
2. Failure	Delete the selected file on Server0	File disappeared from FTP directory listing
3. Recovery	FTP put from PC1 back to Server0	4,414,921-byte transfer completed
4. Verification	Run FTP dir on Server0	File visible again in the directory

Table 3 Failure & Recovery Result

```

ftp>dir

Listing /ftp directory from 192.168.1.10:
0  : asa842-k8.bin                    5571584
1  : asa923-k8.bin                    30468096
2  : c1841-advipservicesk9-mz.124-15.T1.bin  33591768
3  : c1841-ipbase-mz.123-14.T7.bin    13832032
4  : c1841-ipbasek9-mz.124-12.bin    16599160
5  : c1900-universalk9-mz.SPA.155-3.M4a.bin  33591768
6  : c2600-advipservicesk9-mz.124-15.T1.bin  33591768
7  : c2600-i-mz.122-28.bin           5571584
8  : c2600-ipbasek9-mz.124-8.bin      13169700
9  : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin  33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin    5571584
12 : c2800nm-ipbasek9-mz.124-8.bin    15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin  33591768
14 : c2950-i6q412-mz.121-22.EA4.bin    3058048
15 : c2950-i6q412-mz.121-22.EA8.bin    3117390
16 : c2960-lanbase-mz.122-25.SEE1.bin  4670455
17 : c2960-lanbasek9-mz.150-2.SE4.bin   4670455
18 : c3560-advipservicesk9-mz.122-37.SE1.bin  8662192
19 : c3560-advipservicesk9-mz.122-46.SE.bin  10713279
20 : c800-universalk9-mz.SPA.152-4.M4.bin  33591768
21 : c800-universalk9-mz.SPA.154-3.M6a.bin  83029236
22 : cat3k_caa-universalk9.16.03.02.SPA.bin  505532849
23 : cgr1000-universalk9-mz.SPA.154-2.CG    159487552
24 : cgr1000-universalk9-mz.SPA.156-3.CG    184530138
25 : ie9k_iosxe.17.09.04.SPA.bin         596133776
26 : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869
27 : ir800-universalk9-mz.SPA.155-3.M      61750062
28 : ir800-universalk9-mz.SPA.156-3.M      63753767
29 : ir800_yocto-1.7.2.tar              2877440
30 : ir800_yocto-1.7.2_python-2.7.3.tar   6912000
31 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg  685645824
32 : pt1000-i-mz.122-28.bin            5571584
33 : pt3000-i6q412-mz.121-22.EA4.bin    3117390
ftp>

```

Figure 9 Failure simulation


```

ftp>put c2960-lanbase-mz.122-25.FX.bin

Writing file c2960-lanbase-mz.122-25.FX.bin to 192.168.1.10:
File transfer in progress...

[Transfer complete - 4414921 bytes]

4414921 bytes copied in 27.292 secs (4395 bytes/sec)
ftp>dir

Listing /ftp directory from 192.168.1.10:
0 : asa842-k8.bin 5571584
1 : asa923-k8.bin 30468096
2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
3 : c1841-ipbase-mz.123-14.T7.bin 13832032
4 : c1841-ipbasek9-mz.124-12.bin 16599160
5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
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8 : c2600-ipbasek9-mz.124-8.bin 13169700
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10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
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24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
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27 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
28 : ir800-universalk9-mz.SPA.155-3.M 61750062
29 : ir800-universalk9-mz.SPA.156-3.M 63753767
30 : ir800_yocto-1.7.2.tar 2877440
31 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
32 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin 5571584
34 : pt3000-i6q412-mz.121-22.EA4.bin 3117390
ftp>

```

Figure 10 Recovery

9. Testing Results

Test	Expected Result	Observed Result	Status
PC1 -> 192.168.1.10 ping	Replies from Server0	4/4 replies, 0% loss	PASS
PC1 -> office-server ping	Hostname resolves to 192.168.1.10	Replies received from 192.168.1.10	PASS
FTP login	Authenticated file-service access	office account logged in successfully	PASS
Backup download	Selected file copied to PC1	4,414,921 bytes transferred	PASS
Failure simulation	Selected server file removed	File absent from server listing	PASS
Backup restore	Selected file returned to server	4,414,921 bytes transferred and file visible again	PASS

10. Limitations and Recommended Real-World Improvements

- The topology is a single-subnet LAN without a router, firewall, or redundant switch. That is sufficient for the small assignment simulation but not for real office deployment.

- FTP is not equivalent to Windows SMB or Linux Samba. A real implementation should use SMB/Samba with user permissions and controlled shared folders.
- The exercise used a manual FTP get/put cycle to demonstrate backup and recovery. A real backup system should run automatically using a scheduled rsync or robocopy job.
- A production backup should not rely on a client workstation as the only backup target. A separate backup server, NAS, or cloud target should be used, ideally with multiple retention points.
- Printer functionality was represented by the network printer object and static IP address; print-server queuing was not implemented.

11. Conclusion

The office network was successfully simulated in Cisco Packet Tracer. All five end devices were assigned static addresses in the 192.168.1.0/24 network and connected through a 2960-24TT switch. Connectivity tests showed successful communication with 0% packet loss. DNS name resolution was demonstrated by resolving office-server to 192.168.1.10. FTP was used to demonstrate file-service access, and a complete backup, failure, restoration, and verification cycle was executed for a 4,414,921-byte file. The main practical outcome is that the network and recovery workflow can be reproduced from the saved Packet Tracer project, while the limitations of the simulator are clearly documented.